

NOT ALL IS KNOWABLE

Unlike in classical mechanics, in the quantum world some pairs of physical quantities cannot be calculated with certainty. For example, it is impossible to know the exact position and momentum of a particle at the same time; the more precisely one of these quantities is determined, the less precisely the other can be determined. This is known as the uncertainty principle or Heisenberg's uncertainty principle, after German physicist Werner Heisenberg, who discovered the law along with his early framework of quantum mechanics (known as matrix mechanics).

**"We can never
know anything."**

Werner Heisenberg

POSITION

The more certain the position of a particle becomes, the less is known about its momentum.

